MediFind PH: A RAG-Based OTC Medicine Information Assistant Using Pharmaceutical Entity Normalization

Group Apex
ITC-C508 - Deep Learning (NLP)
WW-M2 Group Research Proposal - Revised Version

I. PROPOSED PROJECT TITLE

MediFind PH: A RAG-Based OTC Medicine Information Assistant Using Pharmaceutical Entity Normalization

MediFind PH is a Retrieval-Augmented Generation (RAG) medicine information assistant designed to help consumers retrieve reliable and understandable information about a limited collection of **100-150 selected commonly used over-the-counter (OTC) medicines in the Philippines**. The system focuses on medicine identification, generic and brand names, active ingredients, strength, dosage form, documented purpose or indication, product or formulation availability when supported by reliable data, published or reference-price information, applicable Senior Citizen/PWD discount information, and the source of the retrieved information.

The project is intentionally limited to OTC medicines to keep the knowledge base and evaluation manageable for a student research project. It will not cover prescription-only medicines, diagnose medical conditions, prescribe medicines, recommend personalized dosages, or advise users to replace, change, or discontinue medication.

II. SELECTED UN SDG AND JUSTIFICATION

A. Selected SDG

Sustainable Development Goal 3 (SDG 3) - Good Health and Well-Being.

B. Relevant SDG Area

The proposed project focuses on access to reliable and understandable information about commonly used OTC medicines. Consumers may already know the name of an OTC medicine from its packaging, a pharmacy, or previous professional instruction but may still need reliable information about its identity, formulation, documented indication, and related product information.

C. Justification

Information about OTC medicines can be distributed across regulatory records, package inserts, patient information leaflets, government pharmaceutical references, price references, and official discount policies. In addition, users and official sources may describe the same medicine differently through brand names, generic names, active ingredients, strengths, dosage forms, abbreviations, or formatting. These differences can make relevant information harder to retrieve and understand.

MediFind PH supports SDG 3 by investigating a more accessible way to retrieve verified OTC medicine information. The study addresses an information-access and information-retrieval problem rather than attempting to provide medical treatment.

D. Connection to RAG

RAG allows the chatbot to retrieve relevant evidence from a controlled collection of verified pharmaceutical sources before generating an answer. This can help ground the chatbot's responses in source documents instead of relying only on the language model's internal knowledge.

III. NARROW PROBLEM STATEMENT

A. Problem Narrowing

Level	Group Output
Broad Domain	Health information
SDG-Related Issue	Access to reliable and understandable OTC medicine information
Target Users	Consumers seeking information about a specific OTC medicine they already have in mind
Medicine Scope	100-150 selected commonly used OTC medicines in the Philippines
Narrow Problem	Difficulty retrieving correct OTC medicine information when user queries and pharmaceutical records use different brand names, generic names, active ingredients, strengths, dosage forms, abbreviations, or formats.
Proposed Chatbot Focus	Reliable retrieval of medicine identity, documented purpose, strength, dosage form, supported availability, reference-price, and applicable discount information for the selected OTC medicine collection.

B. Problem Statement

Consumers may struggle to retrieve accurate and understandable information about commonly used OTC medicines because the same medicine can be represented differently across user queries and pharmaceutical sources. A user may enter a brand name, generic name, active ingredient, strength, dosage form, abbreviation, or conversational expression, while the official source may use a different standardized representation. These differences can reduce the likelihood of retrieving the most relevant medicine record or passage.

To keep the study narrow and manageable, MediFind PH will focus only on **100-150 selected commonly used OTC medicines in the Philippines**. The final medicine list will be established using explicit selection criteria and verified pharmaceutical sources; the study will not claim that the collection represents an objective national ranking unless supporting evidence is obtained.

C. Chatbot Focus and Example Questions

A RAG-based chatbot that retrieves reliable information about selected OTC medicines, including generic or brand name, active ingredient, documented purpose, strength, dosage form, supported product/formulation availability, reference-price information, and applicable discounts.

- "What is the generic name of Biogesic 500 mg?"
- "What is this OTC medicine used for according to the source?"
- "Is this medicine documented as a tablet or capsule?"
- "What strengths or formulations are listed for this medicine?"
- "What reference-price or discount information is available for this medicine?"

IV. PHARMACEUTICAL ENTITY NORMALIZATION

A. What It Means

Pharmaceutical entity normalization is the process of identifying medicine-related entities in a user's query and mapping different ways of writing those entities into a consistent representation that can be matched with the knowledge base. In MediFind PH, the important entities may include **brand name, generic/active ingredient, strength, unit, and dosage form**.

Normalization does not create new medical facts. Its purpose is to make the user's wording more consistent with the way medicine information is stored so that retrieval can find the appropriate record or passage.

B. Why It Is Needed

Users may write the same medicine information in different ways. For example, a query may contain *"Biogesic 500"*, *"Biogesic 500mg tablet"*, or *"paracetamol 500 mg"*, while a source may represent the relevant record as *"Biogesic 500 mg Tablet - Paracetamol"*. Exact wording alone may therefore be insufficient for reliable retrieval.

C. How It Works in MediFind PH

Step	Example	Purpose
1. Receive user query	“Ano generic nung Biogesic na 500?”	Accept a natural-language medicine question.
2. Extract entities	Brand = Biogesic; Strength = 500 mg; Intent = generic/active ingredient	Identify the medicine-related information contained in the query.
3. Normalize entities	Biogesic | Paracetamol | 500 mg | Tablet	Map recognized terms to a consistent medicine record or canonical representation.
4. Retrieve evidence	Search the selected OTC medicine knowledge base	Use the normalized information to retrieve relevant records/passages.
5. Generate grounded answer	LLM uses retrieved evidence and returns the answer with its source	Communicate retrieved information without inventing unsupported pharmaceutical facts.

D. Example of Normalization

User query: "Para saan yung Biogesic na 500?"

Entity extraction: Brand = Biogesic; Strength = 500 mg; Intent = documented purpose/indication.

Normalized representation: Brand = Biogesic; Generic/Active Ingredient = Paracetamol; Strength = 500 mg; Dosage Form = Tablet, if these attributes are supported by the selected verified record.

Retrieval: The normalized representation is used to locate the most relevant OTC medicine record or supporting passage. The language model then explains only the information supported by the retrieved evidence.

V. PRIMARY RESEARCH QUESTION

How effectively can MediFind PH use pharmaceutical entity normalization and RAG to retrieve accurate, relevant, and source-grounded information for 100-150 selected commonly used over-the-counter medicines in the Philippines when users express medicine queries using different names, strengths, dosage forms, and natural-language variations?

The question is focused because it limits the study to a defined OTC medicine collection and a specific retrieval challenge: variation in how users express medicine information. It is measurable because the researchers can prepare controlled test questions and evaluate entity normalization, retrieval success, response correctness, relevance, groundedness, and handling of information that is unavailable in the knowledge base.

VI. INITIAL KNOWLEDGE-BASE DATA SOURCES

Source	Organization	Type	Relevance	Availability	Reliability / Permission
FDA Philippines Verification Portal	FDA Philippines	Verification portal / webpage	Registered OTC medicine/product identity and regulatory information.	Public records to be verified before collection.	Official regulatory source; only accessible and authorized information will be used.

Source	Organization	Type	Relevance	Availability	Reliability / Permission
FDA Package Inserts / Patient Information Leaflets	FDA Philippines / registered product information	PI / PIL	Documented indications, strengths, dosage forms, and other approved product information.	Availability for selected OTC medicines will be checked.	Regulatory product information; document authority/version will be verified.
Philippine Government Pharmaceutical References	DOH and relevant government health agencies	Reference / publication / webpage	Potential standardized information for selected OTC medicines.	Only relevant publicly accessible references will be used.	Government-published information; date and suitability will be verified.
DOH Drug Price Reference Index	Department of Health	Government price reference	Supporting published/reference-price information when the selected OTC medicine appears in the source.	Applicable version will be verified.	Values will be labeled according to their actual price type, not automatically as current retail prices.
Official Senior Citizen/PWD Guidelines	Relevant Philippine government agencies	Law / rules / guideline	Applicable medicine discount information and conditions.	Public government documents.	Official policy sources; applicability will be checked.

VII. INITIAL EVALUATION DATA PLAN

The group will create a preliminary test-question set based on the final verified OTC medicine collection and realistic ways users may ask about those medicines. Expected answers and supporting passages will be established from the verified knowledge base.

A. Intended Question Types

- Exact OTC medicine names and standardized forms.
- Brand-name and generic/active-ingredient queries.
- Strength and dosage-form variations.
- Purpose or documented-indication questions.
- English, Filipino, and Taglish conversational queries.
- Formatting variations, abbreviations, and selected minor misspellings.
- Ambiguous queries where more than one strength or dosage form exists.
- Unanswerable questions whose requested medicine or information is absent from the knowledge base.
- Out-of-scope questions requesting diagnosis, medicine recommendation, individualized dosage, or prescription changes.

B. Establishing Expected Answers

For each answerable question, the group will identify the expected normalized medicine entity and the supporting record or passage from the verified knowledge base. For deliberately unanswerable questions, the expected behavior is to state that verified information could not be found rather than generate unsupported information.

C. Planned Evaluation Measures

Possible measures include entity-normalization accuracy, retrieval Hit@K, Precision@K and Recall@K where appropriate, response correctness, relevance, groundedness, and unanswerable-query handling. The final metric set will depend on the implemented experiment.

VIII. POSSIBLE LANGUAGE MODEL AND JUSTIFICATION

A. Proposed Model

Llama 3.1 8B Instruct is proposed as an initial language-model candidate for the future RAG experiment.

B. Justification and Expected Role

The model is proposed because it is instruction-oriented and can formulate conversational responses from retrieved context. In MediFind PH, it will not be treated as the authoritative source of pharmaceutical facts. Relevant evidence will first be retrieved from the verified OTC medicine knowledge base, after which the model will generate an understandable response grounded in that evidence.

C. Practical Considerations

The final model choice will depend on access, deployment requirements, computational resources, response speed, language support, and compatibility with the experiment. English, Filipino, and Taglish queries may be included in future testing. The proposed model is preliminary and may be replaced if another accessible instruction-following model is more suitable.

IX. PROPOSED RAG PROCESS

Knowledge-base preparation: Verified OTC Pharmaceutical Sources -> Data Extraction and Cleaning -> Pharmaceutical Entity Standardization -> Structured Record/Chunk Preparation -> Embedding Generation -> Vector Database.

Query-time process: User Question -> Intent Detection -> Medicine Entity Extraction -> Pharmaceutical Entity Normalization -> Query Embedding -> Semantic/Hybrid Retrieval -> Top-K Relevant OTC Medicine Information -> Retrieved Context + User Question -> Language Model -> Grounded Medicine Information + Source.

The study will focus on whether this process can reliably retrieve the appropriate information despite variations in how users write the names and attributes of the selected OTC medicines.

X. SAFETY AND SCOPE LIMITATIONS

MediFind PH is an information-retrieval assistant, not a clinical decision-support or prescribing system. It is limited to the selected 100-150 OTC medicines and to information supported by the final verified knowledge base.

- **Within scope:** medicine identity, generic/brand information, active ingredient, strength, dosage form, documented indication, supported formulation availability, reference-price information, applicable discount information, and source.
- **Outside scope:** diagnosis, symptom-based medicine selection, personalized dosage, drug-interaction assessment, prescription changes, medication substitution, and advice to start or stop treatment.
- **Current pharmacy stock:** will not be claimed unless a reliable and authorized current inventory source is obtained.

XI. GROUP REFLECTION

The revised topic is sufficiently narrow and manageable because it no longer attempts to cover medicines in general. MediFind PH will be limited to 100-150 selected commonly used OTC medicines in the Philippines and a defined set of information needs. This allows the group to build a controlled knowledge base and a manageable evaluation question set.

The study also has a clear NLP component. Pharmaceutical entity normalization addresses variations in brand names, generic names, active ingredients, strengths, dosage forms, abbreviations, and natural-language expressions before RAG retrieval. The project can therefore evaluate a specific information-retrieval problem while maintaining clear safety boundaries and a realistic student-project scope.